

FINITE RESIDUE CERTIFICATES FOR THE GAP-THREE EQUATION $A^4 + B^4 = (B + 3)^{13}$: BEAL (4, 4, 13) AT LEVEL 26

DAVID FOX

ABSTRACT. We study the gap-three equation $A^4 + B^4 = (B + 3)^{13}$, of signature (4, 4, 13). Fourth-power residues modulo 16 exclude every $B \not\equiv 14 \pmod{16}$. For a specified list of 352 remaining named rows below two million, successive fourth-power non-residue tests at 53, 29, and 109 exclude 266, 77, and 9 rows. This list is not an exhaustive census of that residue class. A separate recorded cover addresses the 62500 values $B \equiv 14 \pmod{16}$ below 10^6 ; the extension to all B retains an explicit Baker-bound premise. Finite level-26 coefficient and trace certificates are distinguished from modularity and level-lowering inputs. We also retain the bounded v33 quotient-Rees gluing manuscript and state the v38 true-chart certificate with its actual hypotheses. The general Beal common-prime statement is recorded in a forward interface whose common-prime proof remains an explicit input. No implication from the chart certificate to that input is claimed.

1. INTRODUCTION

We study finite residue obstructions for $A^4 + B^4 = (B + 3)^{13}$ and distinguish them from conditional modular and logarithmic arguments. The vendored SageMath 10.7 certificate `Level26/BealLevel26Foundations/sagemath/certs/j0_26_decomposition.json` (v1.3.0), generated by `sagemath/j0_26_decomp_foundation.sage` within that package, reports the elliptic factors $26a$ and $26b$ of $J_0(26)$. Their Cremona models $26a1$ and $26b1$ have coefficients $[1, 0, 1, -5, -8]$ and $[1, -1, 1, -3, 3]$, discriminants -17576 and -1664 , and recorded q -expansion prefixes $a_0, \dots, a_{20}$. The package module `lean/Beal/Foundations/J0_26_Decommp.lean` checks finite typed records and distinct prefixes; it does not identify a scheme-theoretic Jacobian. Displayed BSD rational data 1/3

Lean-source baseline: <https://github.com/DavidFox998/beal-conjecture/tree/02728795e15f6c858f0d360d560f80e4efb252a2>. The level-26 package is vendored under `Level26/BealLevel26Foundations/`. The current cited chart version is v37-true-chart, DOI <https://doi.org/10.5281/zenodo.23120540>. Historical software versions are distinguished from fresh verification below.

and $1/7$ are recorded separately, not asserted here as an analytic L-function construction or a proof of BSD. All cited Lean-source evidence is in `beal-conjecture`; no private repository is required as an import.

The populations in this paper must remain distinct. The 352 named rows are not all $B \equiv 14 \pmod{16}$ below two million. The identity $5983 = 4488 + 5 \cdot 299$ is a recorded count, not an enumerated 5983-element Lean list. The later `allKilled_62500` and `allKilled_1e6` statements concern the separate cutoff 10^6 . `baker_conditional_gap3_full` takes $h_{\text{Baker}} : \text{baker_bound_gap3}$ as a premise. The root `BealFinalData.fullB0Search` is an explicit interface input; differently scoped declarations are not identified silently. The finite named-row result is stated in theorem 1; the separately bounded v33 manuscript is retained in section 19, and the general statement and v38 certificate are presented in section 20.

For the historical coefficient ledger we write $C = B + 3$ and $S = (C^{13} - B^{13})/(C - B)$, as in the recorded Φ_{13} identity. The certificate's SHA256 is `35ea70c995f9aed3ae8e2f44cf231d0b7a3ae606e11ad5646635c47fa522a750`; its Git blob is `78a1a982c43201a0a0bdb503a62786ab2b1ef63c`.

2. TRACK B: DENSITY 5983 ROWS

Constants: S_val , $\Phi_{13} = S_val B(B + 3)$, `hasSmallZsigWitness_s_196` by `decide ZMod 53 / 53*53`, `hasSmallZsigWitness_1500003` by `decide ZMod 443 / 443*443` same hit Step 56 (1500003, 1500006). Pools 50310/50323/50307/50350/50325. No Mathlib density import, equality plus count.

3. TRACK A: $S_2(26)$ DIM 2

`newform_26_a_qexp_100 / newform_26_b_qexp_100 = List.take 500 of qExp_26a1_500 and qExp_26b1_500`. Extends $[1, 0, 1, -5, -8]$ and $[1, -1, 1, -3, 3]$. SHA-locked 101-coeff JSON unchanged `0259fe95...`. Prefix equality `decide`. Locked: $a_{53}(26a1) = 0$, $a_{53}(26b1) = 12$, $a_{443}(26a1) = 21$, $a_{443}(26b1) = -39$.

Displayed structure `Newform` is a pair (level, q -exp), not a Mathlib modular form. `newform_26_a/newform_26_b` have level 26 by `rfl`. `exists_newform_level_26_dim2`: $\exists f, g, \text{level } f = 26 \wedge \text{level } g = 26 \wedge f \neq g$. Proof: use those two witnesses; if $f = g$ then q -expansions agree, so $a_{53}(26a1) = a_{53}(26b1)$, but $0 \neq 12$ by `decide`, contradiction. `#print axioms exists_newform_level_26_dim2` empty. This does *not* inhabit `Chain ExistsNewformLevel2` ($S_2(\Gamma_0(2))$ displayed dimension 0).

4. FREY TRACES

Displayed Weierstrass $y^2 = x(x - B^4)(x + C^4)$ Euler count `decide` not `native_decide`: $a_{53}(E_{196}) = -2$, $a_{443}(E_{1500003}) = 24$. The definitions are in `Level26/BealLevel26Foundations/lean/BealLevel26Foundations/Beal/FullProof/Level26_Newforms.lean`, lines 235–278 at the source baseline. The equation’s Beal Frey cubic $y^2 = x(x - A^4)(x + B^4)$ is defined separately. No identification of these models or transport of their traces is used without a proved bridge.

5. FREY MOD 13 IRREDUCIBLE VIA $p = 53, 443$ TRACES -2 AND 24

`frey_mod13_irreducible` at $B = 196$ or $B = 1500003$ is the displayed Int-mod-13 miss: $(-2)\%13 \neq 0\%13, 12\%13$ and $24\%13 \neq 21\%13, (-39)\%13$, via `irreducible_of_trace_mismatch` applied to `level26_a_eliminated_by_53 / level26_b_eliminated_by_443`. This is *not* Mazur, not Borel image, and not a Mathlib residual representation. `#print axioms frey_mod13_irreducible` empty. `frey_modular` stays `Classical.em`. `level_lowering_to_26` stays an uninhabited `Prop`; the inhabited pack remains `ribet_mazur_pack_q_13_level_26` via `hK.left / hK.right`.

6. KRAUS MOD 13 v8.69.0 93a0c1d

`level26_a_eliminated_by_53`: $(-2)\%13 \neq 0\%13 \wedge (-2)\%13 \neq 12\%13$ (`decide`). `level26_b_eliminated_by_443`: $24\%13 \neq 21\%13 \wedge 24\%13 \neq (-39)\%13$ (`decide`). `kraus_elimination_q_13_level_26`: conjunction via `pair`. Theorem name not Lean type, combined statement uses inequalities.

7. RIBET-MAZUR PACK v8.71.0

`kraus_elimination_q_13_level_26_density` for $C = B + 3$ and Density $\Phi_{13}B = S_valBC$, $B = 196$ and $B = 1500003$ branches are v8.69.0 misses (exact `level26_a_eliminated_by_53 / level26_b_eliminated_by_443`). `ribet_mazur_pack_q_13_level_26` implication to $\forall B, (B = 196 \rightarrow \dots) \wedge (B = 1500003 \rightarrow \dots)$ via `hK.left / hK.right`. Not $\forall B$ modular contradiction. Density/Step `kraus_elimination_q_13_level_26` stays uninhabited $\forall$. Displayed `exists_newform_level_26_dim2` inhabited. Chain `ExistsNewformLevel2` stays $0 \neq 0$. No new Beal $\forall$. `beal_4_13_13_gap3_B_le_2M_eliminated` stays `Classical.em`. `zsig_density_links_to_kraus` list equality $+4488 + 5 * 299 = 5983$. `#print axioms ribet_mazur_pack_q_13_level_26` empty, density theorem `[propext, Quot.sound]` inside allowed.

8. REPRODUCIBILITY FOR MCOM

The historical scaffold record reports successful builds of `BealLevel26Foundations` and `BealLevel26FoundationsFullProof`, and a passing `scripts/verify-scaffold.sh`. These are historical reports, not independent replay results from the present manuscript revision. The axiom footprint must be checked for each cited declaration; it is not a repository-wide certificate. Fresh-check commands and the local limitations are given in appendix A.

9. MOD16

Fourth powers modulo 16 are 0 or 1. Odd thirteenth powers satisfy $c^{13} \equiv c \pmod{16}$; even thirteenth powers with exponent ≥ 4 are 0.

$B \bmod 16$	B^4	$(B+3)^{13}$	$A^4 + B^4 \equiv (B+3)^{13}?$
odd	1	0	no ($A^4 + 1 \equiv 0$)
0, 2, 4, 6, 8, 10, 12	0	3, 5, 7, 9, 11, 13, 15	no
14	0	1	open ($A^4 \equiv 1$)

Thus 15 of 16 residue classes are eliminated by `decide` on residues (kernel `Nat.pow`). `beal_4_13_13_gap3_B_le_2M_eliminated_mod16` is that kill on $B \leq 2000000$ with $B \not\equiv 14$; `#print axioms` is empty. The named-row count $4488 + 5 * 299 = 5983$ is the existing density identity, not an enumeration of 5983 numerals. $B = 196$ ($4 \not\equiv 14$) and $B = 1500003$ (odd) are instances. Remaining $B \equiv 14 \pmod{16}$ are not killed by residues. The displayed list `density_5983_list` = [196, 1500003] has no such member.

10. $B \equiv 14$ KRAUS SWEEP

Step60 names 352 rows with $B \equiv 14 \pmod{16}$ (not ~ 374 of a 5983-numeral dump). Displayed Frey traces $a_{53}(E_B) = -\sum \chi(x(x - B^4)(x + (B+3)^4))$ on $\mathbb{F}_{53}$ land in $\{-2, -10, 1, 6, 14\}$ and all miss locked $a_{53}(26a1) = 0$ and $a_{53}(26b1) = 12$ at $\ell = 13$.

B	$a_{53}(E_B)$	vs 0/12	note
6046	6	$6 \not\equiv 0, 12$	Euler decide
18526	-2	$-2 \not\equiv 0, 12$	Euler decide
350 other named rows	$\{-2, -10, 1, 6, 14\}$	all miss	recorded

That is an Int-mod-13 miss pack, not Ribet and not $\neg \exists A$. `beal_4_13_13_gap3_B_le_2M_eliminated_B14_kraus` is the vacuous implication on the displayed two-element list (neither member is $\equiv 14$). `beal_4_13_13_gap3_B_le_2M_eliminated_full` is those two rows via the v8.76.0 residues. `#print axioms` is empty on both. The v8.69.0–v8.74.0 pack (-2 vs $0/12$, 24 vs $21/-39$) stays recorded.

11. $B \equiv 14$ LIST AND DISPLAYED 2-TORSION

`step60_b14_list` is the 352 named Step60 numerals, all $\equiv 14 \pmod{16}$. `b14_witnesses` is the matching $(B, 53, a_{53})$ list with $a_{53} \in \{-2, -10, 1, 6, 14\}$. The displayed cubic $x(x - B^4)(x + (B + 3)^4)$ has roots $0, B^4, -(B + 3)^4$ (`frey_modular_B14`, axiom `propext`). That is not Wiles and not BCDT. `level_lowering_to_26_B14` stays a Prop: the recorded pack is a miss, not a congruence $a_E \equiv a_f \pmod{13}$. `beal_4_13_13_gap3_B_le_2M_eliminated_B14_honest` stays the uninhabited $\forall$ on those 352 rows. `beal_4_13_13_gap3_B_le_2M_eliminated_full_honest` is the displayed two-element list via mod 16; `#print axioms` empty.

12. $B \equiv 14$ HONEST: BEAL FREY CUBIC, NOT BCDT

The Beal Frey cubic is the integer polynomial $x(x - A^4)(x + B^4)$ with roots $0, A^4, -B^4$ (`beal_frey_curve / beal_frey_has_full_2_torsion`, axiom `propext`). That is not the displayed cubic $x(x - B^4)(x + (B + 3)^4)$ of the previous section (`displayed_cubic_ne_beal_frey_example`, empty axioms). If $A^4 + B^4 = (B + 3)^{13}$, the last factor rewrites as $x - A^4 + (B + 3)^{13}$ (`beal_frey_rewrite`, axiom `propext`). The 352 named $B \equiv 14$ rows still record displayed-cubic traces $a_{53} \in \{-2, -10, 1, 6, 14\}$, all missing locked 0 and 12 at $\ell = 13$. Those Euler counts are not a_{53} of the Beal Frey cubic. `beal_frey_modular`, `beal_frey_mod13_irreducible_B14`, and `level_lowering_to_26_B14_honest` stay Props: there is no BCDT theorem in this graph, `Classical.choice` is not Wiles, and the Ribet–Mazur pack is a miss pair, not a congruence $a_E \equiv a_f \pmod{13}$. `beal_4_13_13_gap3_B_le_2M_eliminated_B14_honest` stays the uninhabited $\forall$ on those 352 rows. `beal_4_13_13_gap3_B_le_2M_eliminated_5983_honest` is the uninhabited conjunction of the two-element mod-16 kill with that Prop. The $5983 = 4488 + 5 \cdot 299$ identity remains `zsig_density_2M_stats`. Allowed axioms on the inhabited cubic lemmas are in `{propext, Quot.sound, Classical.choice}`; the cubic 2-torsion and rewrite use `propext` only. Not `Classical.em`.

13. GAP-3 GCD AND DISPLAYED MISS; 86-ROW ARROWS STAY PROPS

`gcd_B_Bplus3_dvd_three` inhabits $\gcd(B, B + 3) \mid 3$ (axiom `propext`). That is not a conductor computation. `irreducible_86` is `IrreducibleMod13Trace` of Euler a_{53} , the same integer miss as `beal_frey_a53_miss_86`, not Mazur and not a Mathlib residual representation. `conductor_86`, `beal_frey_modular_allowed`, `level_lowering_86`, and `beal_4_13_13_gap3_B_le_2M_eliminated_86` stay uninhabited

Props. `Classical.choice` is not Wiles and not Ribet. A custom axiom `BCDT_B14` would print as `BCDT_B14`. The Zsigmondy identity $5983 = 4488 + 5 \cdot 299$ does not prove $N \mid 2^5 \cdot 3 \cdot 13$: survivor 63982 = $2 \cdot 31991$ is already not $\{2, 3, 13\}$ -smooth. `beal_frey_curve` is an integer polynomial, so there is no Mathlib `.conductor`.

14. $B \equiv 14$ EULER a_{53} VALUE SET ON THE 86 SURVIVORS

`a53_res_in_table` records that every fourth-power pair has Euler a_{53} in $\{-10, -2, 0, 1, 6, 14\}$ (axioms `propext`, `Quot.sound`). Under $A^4 + B^4 = (B + 3)^{13}$ the singular cell $(0, 0)$ is excluded, so `beal_frey_a53_value_of_eq` and `beal_frey_a53_miss_86` inhabit $a_{53} \in \{-10, -2, 1, 6, 14\}$ together with the miss of locked 0 and 12 at $\ell = 13$ on the 86 named rows that survive the mod-53 fourth-power test (axioms `propext`, `Quot.sound`). There is no Mathlib $a_{53}(\text{beal_frey_curve})$. That is not $\neg \exists A$ for those 86. `beal_4_13_13_gap3_B_le_2M_eliminated_B14_honest` stays the uninhabited $\forall$ on all 352 rows. Not `BCDT` and not Ribet. Not `Classical.em`.

15. $B \equiv 14$ A-MOD-53 SEARCH, NOT `BCDT`

Fourth powers modulo 53 are the 14 residues $\{0, 1, 10, 13, 15, 16, 24, 28, 36, 42, 44, 46, 47, 49\}$. Euler a_{53} of the Beal Frey cubic $y^2 = x(x - A^4)(x + B^4)$ on $\mathbb{F}_{53}$ is searched on that 14×14 table (`a53_res_miss_or_zero`, empty axioms). Values lie in $\{-10, -2, 0, 1, 6, 14\}$. The cell $(A^4, B^4) \equiv (0, 0)$ is the singular cubic $y^2 = x^3$ and gives $a_{53} = 0$. The other five miss locked 0 and 12: $-10 \equiv 3$, $-2 \equiv 11$, $1 \equiv 1$, $6 \equiv 6$, $14 \equiv 1 \pmod{13}$. Under $A^4 + B^4 = (B + 3)^{13}$ both residues cannot be 0 (that would force $53 \mid 3$), so `beal_frey_a53_miss_B14` inhabits the miss on the 352 named rows (axioms `propext`, `Quot.sound`). A custom axiom `BCDT_B14` would print as `BCDT_B14`, not as `Classical.choice`. `beal_frey_modular` and `beal_4_13_13_gap3_B_le_2M_eliminated_B14_honest` stay Props. Not `Classical.em`.

16. $B \equiv 14$ FOURTH-POWER RESIDUE COVER

Fourth powers modulo 53 are the 14 residues $\{0, 1, 10, 13, 15, 16, 24, 28, 36, 42, 44, 46, 47, 49\}$. If $(B + 3)^{13} - B^4$ is not a fourth power in $\mathbb{F}_{53}$, there is no A with $A^4 + B^4 = (B + 3)^{13}$ (`direct_mod53_kill`). On the 352 named $B \equiv 14$ rows this kills 266; `#print axioms` empty. The count is 266, not 287.

The remaining 86 die at further primes (`direct_modq`). $q = 29$ kills 77. $q = 109$ kills the remaining 9: 214430, 312798, 477854, 603358, 813118, 863694, 1136270, 1287998, 1420830. $q = 17$ is the named

demo on those 86 mod-53 survivors: fourth powers $\{0, 1, 4, 13, 16\}$, 57 killed (`direct_mod17_kill`). `beal_4_13_13_gap3_B_le_2M_eliminated_86_modq` inhabits $\neg\exists A$ on all 86; `#print axioms` empty.

`beal_4_13_13_gap3_B_le_2M_eliminated_B14_full` composes those three residue tests and inhabits $\forall B \in \text{step60_b14_list}, \neg\exists A$ ($266 + 77 + 9 = 352$); `#print axioms` empty. The user's `by_cases/decide` draft is rewritten as that empty-axiom cover. This name is not KrausB14's `beal_4_13_13_gap3_B_le_2M_eliminated_full`, which remains the two-element mod-16 kill of $[196, 1500003]$ (`full_honest`).

`beal_frey_a53_miss_86` still records Euler $a_{53} \in \{-10, -2, 1, 6, 14\}$ missing 0, 12 at $\ell = 13$ (`axioms propext`, `Quot.sound`). That pack is an Euler miss, not Ribet and not the Beal negation.

`conductor_86` stays a Prop. The first survivor identity $63982 = 2 \cdot 31991$ is empty. `gcd_B_Bplus3_dvd_three` is only $\gcd(B, B + 3) \mid 3$ (axiom `propext`), not a conductor.

`beal_4_13_13_gap3_B_le_2M_eliminated_B14_honest` (the 352-row pack $\forall$), `density/step kraus_elimination_q_13_level_26`, and `Chain ExistsNewformLevel12` ($0 \neq 0$) stay uninhabited Props intentionally. The Zsigmondy count $5983 = 4488 + 5 \cdot 299$ is unchanged (`zsig_density_2M_stats`). Not BCDT and not Ribet. Not `Classical.em`.

17. MCOM SUBMISSION (v8.86.0 RESIDUE THEOREM)

Theorem 1 (gap 3 at $B \leq 2M$). *The empty-axiom residue cover of v8.86.0-paper-B14-final-DOI eliminates the named $B \leq 2M$ gap-3 rows: if $B \leq 2000000$ and $B \not\equiv 14 \pmod{16}$, there is no A with $A^4 + B^4 = (B + 3)^{13}$ (`beal_4_13_13_gap3_B_le_2M_eliminated_mod16`); if B is one of the 352 named $B \equiv 14$ rows, there is no A via fourth-power non-residues modulo 53, 29, and 109 (`beal_4_13_13_gap3_B_le_2M_eliminated_B14_full`). `#print axioms` empty on both.*

The $5983 = 4488 + 5 \cdot 299$ census is $B \leq 2M$ density, not a $> 2M$ bound. The identity $63982 = 2 \cdot 31991$ shows that the displayed survivor is not $\{2, 3, 13\}$ -smooth, so `conductor_86`, `B14_honest`, `density/step kraus_elimination_q_13_level_26`, and `Chain ExistsNewformLevel12` ($0 \neq 0$) remain Props intentionally. `frey_conductor_general` gives $|\Delta| = 2^4 \cdot N_0$ with $N_0 \mid (ABC)^{26} \cdot 13$ (`axioms propext`, `Quot.sound`); that is not Tate $N(E)$. Tate Step 2 at an odd prime of ABC gives local exponent 1 when the bases are pairwise coprime (`tate_odd_prime_exponent_one`); $v_2(c_4) \geq 4$ so Step 2 does not finish at 2. Tate Steps 6–7 at 2 give `tate_2adic_exponent_1e5` ($v_2(c_6)$ from A, B parity; local exponent ≤ 5). The defined packed conductor `tateConductor` is $2^{f_2} \text{rad}(AB(B + 3)) \cdot 13$ with $f_2 = \text{conductorExponentTate2}$; `tate_con`

`ductor_bound_rhs` is the same `Nat`. `frey_tate_conductor_inhabited` proves `tateConductor` $\mid 2^5 \text{rad} \cdot 13$ with $v_q \leq 1$ odd and $v_2 \leq 5$. This is a packed displayed `Nat`, not Mathlib Tate $N(E)$. `conductor_86` stays the stronger uninhabited $N \mid 2^5 \cdot 3 \cdot 13$ ($63982 = 2 \cdot 31991$ proves that old claim is false; we use $2^5 \text{rad} \cdot 13$).

$B > 2M$ gap 3 is the open Prop `baker_bound_gap3` (explicit Matveev for (4, 4, 13) gap3, not general Bugeaud; Mathlib 4.12 has no Matveev / BMS Table 1). `matveev_explicit_gap3` turns that Prop into $A^4 + B^4 = (B + 3)^{13} \Rightarrow B \leq B_0$. `matveev_no_solution_A_le_B` is inhabited: $A \leq B$ never solves gap 3. It is not closable from the 5983 census. Conditionally, `baker_conditional_gap3_full` gives $\forall B$ from that premise: $B \leq B_0$ by the expanded residue cover (mod 16 plus fourth powers at 53, 29, 109 and 17, 5, 7, 11, 13, 19, 23, 31, 37); the 62500 values $B \equiv 14 \pmod{16}$ up to 10^6 are the 25 kernel slices `allKilled_chunk_0`–`allKilled_chunk_24`, conjoined as `allKilled_62500` and `allKilled_1e6`; $B > B_0$ by Baker. `exists_zsig_q_gt_2M_dvd_C13_sub_B4` needs Baker, not Zsig, and stays a Prop. `sorry` is not used. The level-26 modular-elimination pack `beal_44_13_level_26_modular_elimination` takes five transparent cert structures (`J0DecompositionCert_26`, `MwrankCertificate_26`, `FormalImmersionCert_26`, `FreyCurveCert`, `LevelLoweringCertificate_26`; soundness is a computable existential) and the still-uninhabited `baker_bound_gap3`. The Frey model is $Y^2 = X(X - A^4)(X + B^4)$ with packed conductor `tateConductor` $= 2^{f_2} \text{rad}(AB(B + 3)) \cdot 13$, $f_2 \leq 5$, not $2^5 \cdot 3 \cdot 13$ ($63982 = 2 \cdot 31991$). `kraus_a53_elimination` is the kernel decide that $a_{53} \in \{-10, -2, 1, 6, 14\}$ misses $S_2(26)$ traces $\{0, 12\}$. Tate and Baker files are unchanged.

`v24.x` final summary. Versions `v24.0.0`–`v24.3.2` build the displayed Matveev kernel without claiming $|\Lambda| > \exp(C)$. The chain of version DOIs is 10.5281/zenodo.22730408, 22730460, 22730548, 22730776, 22730846, 22731054, 22731160, 22731508, 22731634, 22731759, 22731966, 22732045, with `v24.4.0` 10.5281/zenodo.22732209. On a gap-3 solution, `matveev_v24x_summary` packs $|\Lambda| = \log(1 + B^4/A^4) \leq B^4/A^4$ together with $0 < \exp(C_{\text{exp_bound}}) < 10^{-12} < 1$ and $C_{\text{exp_bound}} = -(\text{height}_{B_0} \log \text{height}_{B_0})$, where

$$\begin{aligned} \text{height}_{B_0} &= 104382751019310000000 \\ &= C_1^{\text{floor}} \cdot 30^6. \end{aligned}$$

`matveev_v24x_conditional_summary` is the implication from the still-uninhabited `matveev_inequality_real_target`. `bugeaud_v24x_final_link` restates the conditional LLL shape. This is the honest final of

v24.x, not v25: `baker_bound_gap3` stays a def Prop until Mathlib has Matveev 2000 Thm 1.4. The closing software record is [1].

18. WHAT REMAINS

15/16 of $B \leq 2M$ is eliminated by mod-16 residues. The two displayed named rows [196, 1500003] are eliminated that way (`full_honest`). All 352 named $B \equiv 14$ rows are eliminated by fourth-power non-residues (`beal_4_13_13_gap3_B_le_2M_eliminated_B14_full`). Those 86 that survive mod 53 still have an Euler a_{53} in $\{-10, -2, 1, 6, 14\}$ that misses 0, 12 under the Beal equation (`beal_frey_a53_miss_86`); that pack is not Ribet. `conductor_86` and `level_lowering_86` stay Props. `B14_honest` stays the uninhabited pack $\forall$. `Classical.choice` is not BCDT. The old `beal_4_13_13_gap3_B_le_2M_eliminated` stays `Classical.em`. Density from the named-row count to ∞ , full modular contradiction, and Beal $\forall$ still open. $B > 2M$ is the Baker/Bugeaud Prop `baker_bound_gap3`, not a Zsigmondy theorem. Track beyond 2M Step61+ continues the same `smallZsigPrimes` pattern and does not close the Baker bound.

19. WORKING v33 PAPER - BOUNDED QUOTIENT-REES GLUING (INCORPORATED FROM REPLIT STANDALONE)

The build source was `papers/main.tex`, generated PDF `main.pdf`. The full source is archived at `paper/v33-standalone/main.tex` and the full PDF at `paper/v33-standalone/main.pdf`. This section is that manuscript's body, not a truncation. It is a bounded description of the quotient-Rees three-open gluing and the actual blow-up's generic-fibre comparison. It is not a proof of the general Beal conjecture, a global identification of the actual special-fibre pullback with the quotient Rees Proj, or a degree-one / nonvanishing certificate for $\overline{2t}$.

The archived manuscript's deposit-planning language is historical. The current repository audit records a published v33 GitHub release and version DOI 10.5281/zenodo.23054568; see `Beal/AUDIT_v33_HONESTY.md`, lines 79–88 and 120–124. This revision mints no DOI. The current cited chart version remains v37-true-chart, [2]; the old deposit language does not replace that citation. The level-26 foundations work mirror is not the v33 release source. See `Beal/AUDIT_v33_HONESTY.md`, `Beal/ZENODO_DEPOSIT_v33.md`, and `Beal/RELEASE_NOTES_v33.md`.

This incorporation does not change the beal-v38 EQUIV:3 chain `ddfb2642` $\rightarrow$ `e466e5a` $\rightarrow$ `792b3f8`, the freeze tag `beal-v38-freeze-Y-axis-glue` object `437b4c85` at `c48b1bd3`, or the v37 citation <https://doi.org/10.5281/zenodo.23120540>.

19.1. Abstract of the v33 manuscript. The following abstract is retained from the archived working manuscript. Its release and web-hook statements describe that manuscript’s historical planning stage, not the current deposit status recorded above.

This working v33 manuscript records a three-open `OpenCover` of the *quotient* Rees scheme $\text{Proj}(\text{ReesMod2})$ by $D_+(\overline{2t})$, $D_+(\overline{Xt})$, and $D_+(\overline{Yt})$, with affine restriction squares for every ordered pair over abstract product-open pullbacks. `CompatChart2t` (191 lines) identifies the principal divided-chart kernel (F) and proves `chart_eq_quotient_2t`; `CompatChartXt` and `CompatChartYt` (105 lines each) prove $J_X = (G_X : X^\infty)$ and $J_Y = (G_Y : Y^\infty)$ and their quotient-induced chart identifications. Six directed polynomial-to-abstract restrictions satisfy $\text{res}_{\text{poly}} = \text{res}_{\text{actual}} \circ \psi_\bullet$. `SpecialFibreGluing` (167 lines) checks six composed `Spec.map` compatibilities and the abstract triple cocycle, giving

$$\text{Proj}(\text{ReesMod2}) \cong \text{Glue}(S_{2t}/(2), S_{Xt}/(2), S_{Yt}/(2)).$$

Product overlaps are abstract pullbacks, not polynomial presentations. For $y^2 + xy = x^3 + 8$, the reduced $2t$ chart is $V^2 + UV = 0$; the point $U = V = 0$ lies only in $D_+(\overline{2t})$ (an illustration, not a separately named Lean theorem). `TateEvenBranchGenericFibre` (59 lines) packages the *actual* blow-up comparison $\text{Bl}_I[1/2] \cong \text{Spec}(R_Z[1/2])$ with $I[1/2] = \mathbb{T}$; the existing 2,249-line chart file is untouched. The seven targeted `lake env lean` direct checks previously passed with printed axioms only `propext`, `Classical.choice`, and `Quot.sound`; the scoped `sorry` scan and `git diff --check` are clean. A standalone direct check of the large special-fibre file timed out without a diagnostic and a full `lake build BealGeneral` remains unverified. We do *not* claim a global isomorphism from the actual special-fibre pullback to the quotient Rees Proj , or a degree-one quotient formula or nonvanishing theorem for $\overline{2t}$. This is not a general Beal proof. The intended release and Zenodo source is `beal-conjecture`, not its level-26 work mirror; its active GitHub-side Zenodo `release` webhook does not yet establish delivery, a deposit, or a DOI. A DOI may be minted only after an authorized GitHub Release v33 and subsequent Zenodo verification.

19.2. Scope and charts. Write $R_Z = \mathbb{Z}_2[X, Y]/(f)$ for the translated even-branch surface under the stated nodal and coefficient-divisibility hypotheses, $I = (2, X, Y)$, and $\text{ReesMod2} = \text{Rees}(I)/(2)$, where (2) is generated by the degree-zero base scalar, *not* by $2t$. The integral divided chart has $f(2U, 2V) = 4F$; its principal relation is (F) . The Xt and Yt charts use saturated graph ideals $(G_X : X^\infty)$ and $(G_Y : Y^\infty)$ rather than the unsaturated graph equations. Quotient-induced chart

maps connect these integral calculations to the three reduced chart spectra. These assertions concern the charts under their hypotheses, not an unconditional formula for every Weierstrass model.

19.3. The checked gluing boundary. The three reduced generator opens cover $\text{Proj}(\text{ReesMod2})$. Ordered affine restriction squares live over product opens; the six distinct directed pairs obey the polynomial-to-abstract restriction identities. The polynomial spectra are literal cover objects, but their *overlaps* are abstract product pullbacks. Six Spec.map squares and a triple cocycle yield the global isomorphism displayed in the abstract. No polynomial ring presentation of a product overlap is supplied or inferred.

The illustrative $y^2 + xy = x^3 + 8$ specialization has $f(2U, 2V) = 4(V^2 + UV - 2U^3 - 2)$, so its reduced divided-chart equation is $V^2 + UV = 0$. The point $(U, V) = (0, 0)$ gives direction $[2t : Xt : Yt] = [1 : 0 : 0]$ and belongs to the $2t$ open alone; this is an example, not a global classification theorem.

19.4. Generic fibre and unresolved special-fibre comparison. After inverting 2, the centre I becomes the unit ideal. The packaged generic-fibre isomorphism concerns the *actual* blow-up and the localized translated surface over $D(2)$; it does not identify an integral chart with its generic fibre. In contrast, the checked special-fibre gluing concerns the *quotient Rees Proj*. Chartwise base-change comparisons do not yet give an isomorphism of that Proj with the actual scheme-theoretic special-fibre pullback. Nor is $(\text{ReesMod2})_1 \cong (I/2I)t$ or $\overline{2t} \neq 0$ certified here. These missing steps are not used as premises of the quotient- Proj gluing.

19.5. Verification and deposit route. The seven direct Lean checks and their axiom reports are recorded in `Beal/RELEASE_NOTES_v33.md` and `Beal/AUDIT_v33_HONESTY.md`. They are not a verified full-target build: the standalone large special-fibre source check timed out, although the passing modules imported its compiled declarations. The existing `TateEvenBranch.lean` (2,249 lines) passed its separate direct check and was not refactored.

`beal-conjecture` is the whole-project main audit and intended GitHub-release source. The level-26 scaffold's Lean evidence is cited through the vendored `Level26/BealLevel26Foundations/lean/` tree at the immutable `beal-conjecture` baseline, not through an external code pointer. Keep that scaffold separate from an inhabited global Beal result. The archived manuscript said, “Neither a tag nor a release

nor a v33 DOI exists yet.” That sentence is historical: the current audit records the v33 release and DOI. This revision performs no release, deposit, DOI mint, or mirror push.

19.6. PDF printing. Build from `paper/` with `latexmk -pdf mcom-draft.tex` or `tectonic mcom-draft.tex`. The generated file is `paper/mcom-draft.pdf`. The single source directory is `paper/`.

20. GENERAL BEAL STATEMENT AND THE V38 TRUE-CHART CERTIFICATE

In this section Y, S denote the cusp-chart variables, independently of the coefficient-ledger notation in section 1. Write $\overline{N} = \text{closedYtToCusp } N$ for the cusp class.

Theorem 2 (True-chart presentation and E_i certificate (v38)). *Let*

$$R = \mathbb{F}_2[a, b][X, Y, U, V],$$

$$I_{\text{true}} = (XU, Y - XV, Y^2 - X^3, U + X^2 + XV^2).$$

The true model R/I_{true} is ring-isomorphic to the fixed chart `chart_Dplus_Xt_ring valuationOneCurve 0 0` via `chart_Dplus_Xt_true_presentation`. Its map $\iota = \text{chartOfModelTrue}$ is injective. For $N : \text{modelYtChart_fixed}$ with `chartOfModelTrueY\_fixed` $N = 0$, the certificate supplies a cusp representative

$$\overline{N} = A + YB + Y^2C + Y^3(E_0 + SE_1 + S^2E_2).$$

Here A, B, C have Y -degree zero, and E_0, E_1, E_2 have S -degree zero. The high part $Y^3(E_0 + SE_1 + S^2E_2)$ has S -degree at most two. Its chart image equals that of the low part $A + YB + Y^2C$; writing the high-part image as H_N , it satisfies $(1 + YS^3)H_N = 0$. The recorded nonvanishing and annihilator conjuncts are retained: no vanishing of B or E_i is inferred.

Proof. The ring presentation is `RingEquiv.ofBijective` applied to the existing injectivity and surjectivity proofs in `Beal/MathlibMissing/ChartInjective.lean`, lines 3162–3176. The v38 conjunction in `Beal/MathlibMissing/ChartTrueEquiv.lean`, lines 261–338, assembles `chartOfModelTrue_injective`, `chartTrueEquiv_inj_from_Ei_constraint N hN`, and `chartTrueEquiv_annihilator_presentations`. The statement above displays their chart-presentation and normal-form components, not a new arithmetic implication. $\square$

Nonvanishing and overlap invariants. The $B = 1$ example remains nonzero outside the cusp; Y^3 is nonzero and outside the cusp ideal. In the centre-power assertion, I is the centre ideal of the fixed 2-adic model, not the characteristic-two ideal I_{true} . In particular,

$$\text{centreNormalPoly}(X^3 - 1)0 \notin I^2, \quad \text{centreAlphaBound}20 = 1.$$

The constant coefficient of $X^3 - 1$ is a unit in the 2-adic coefficient ring, incompatible with that required centre-power bound. The relation $X^2(X + V^2)z = 0$ is distinct from $B = 0$; $X + V^2$ stays outside the cusp ideal with nonzero image. On the overlap, $SV = 1$, $Y = XV$, and $X = YS$. With $\text{ann}(f)$ denoting the annihilator of the principal ideal generated by f , the recorded comparisons are

$$\begin{aligned} \text{ann}(1 + YS^3) &= \text{ann}(X + V^2), \\ \text{ann}(X^2) &= \text{ann}(Y^2), \\ \text{ann}(1 + YS^3) &\neq \text{ann}(X^2). \end{aligned}$$

The distinguishing witness is Y^2 : it annihilates the factor $1 + YS^3$, not the power X^2 . These are annihilator presentations. The source supplies no ring map $D_+(Xt) \rightarrow \text{overlap}$; no such arrow is inferred. The kernel hypothesis on N refers to its defined chart map, not to an invented map into the overlap.

The recorded general statement. This records the general Beal conjecture succinctly: for positive bases and exponents at least three, an equality $x^p + y^q = z^r$ entails a common prime divisor. The formal forward interface has an explicit input $D : \text{BealFinalData}$, including the field $D.\text{bealTheorem.commonPrime}$:

$$\begin{aligned} D : \text{BealFinalData}, \quad x, y, z > 0, \quad p, q, r \geq 3, \\ x^p + y^q = z^r \implies \exists \ell \in \mathbb{N}, \quad \ell \text{ prime}, \quad \ell \mid x, \ell \mid y, \ell \mid z. \end{aligned}$$

`beal_main_theorem_forward` returns that common-prime input field in `BealFinal/Main.lean`, lines 74–80. Theorem 2 displays chart injectivity and the accompanying E_i and annihilator data; it does not supply the common-prime input. The v33 result in section 19 remains separately bounded.

20.1. Immutable provenance.

Tag or version	Object	Target or citation
<code>beal-v38-freeze-Y-axis-glue</code>	437b4c85	c48b1bd3
<code>beal-v38-equiv-Ei-constrain</code>	c2f530ab	792b3f8
<code>t</code>		
<code>v37-true-chart</code>	Version DOI	10.5281/zenodo.23120540

The proof record is `ddfb2642 → e466e5a → 792b3f8`: the E_i constraint associated with `025b34c2`, then the nonzero $B = 1$ example, then `EQUIV:3`. The code baseline is `02728795e15f6c858f0d360d560f80e4efb252a2`; the manuscript revision starts from PR28 at `75b8a3a5`. These commit and tag-object identifiers are not new DOIs.

APPENDIX A. EXACT-TARGET REPRODUCIBILITY AND SOURCE LEDGER

Retain Lean/Mathlib 4.12.0 and the checked-in manifest. Do not run `lake update` for this pinned replay. From the repository root, run each target separately and retain its complete log and actual exit status:

```
lake --version
lake exe cache get
lake build BealMathlibMissing
lake build BealEven
```

After successful target builds, directly check `Beal/MathlibMissing/ChartTrueEquiv.lean`; its seven `#print axioms` commands are at lines 340–346. The repository reports `[propext, Classical.choice, Quot.sound]` for those declarations. Report any `sorryAx` in the replay. This is scoped, not an axiom certificate for every module. Other modules contain explicit declarations `darmon_merel_4413_axiom`, `frey_modular_13`, and `ribet_level_lowering_26`; their role must be assessed for each cited theorem.

The manuscript review’s independent cache-setup attempts exited 137 before either requested target ran; the environment recorded two OOM kills. A lexical scan found no active `sorry`, `admit`, or `sorryAx` tokens in the resolved local import closures: 57 files for the 18 `BealMathlibMissing` roots and 39 files for the one `BealEven` root. External dependencies were excluded. That scan is not elaboration or a printed axiom report. The inspected workflow builds other Matveev targets; its status does not certify these two exact targets. The earlier PR28 PDF built to 11 searchable pages; pagination changes in this revision are allowed for readability.

Repository-only Lean evidence. Use immutable paths within `DavidFox998/beal-conjecture`:

- Finite trace data: `Level26/BealLevel26Foundations/lean/BealLevel26Foundations/Beal/FullProof/Level26_Newforms.lean`.
- Named residue cover and conditional bound: `BealFreyMod53Kill.lean`, `BealFreyModQKill.lean`, and `BealGap3BakerUpperBound.lean` in that same `FullProof/` directory.

- Bounded gluing: `Beal/Beal.General/SpecialFibreGluing.lean`.
- True chart and v38 data: `Beal/MathlibMissing/ChartInjective.lean` and `Beal/MathlibMissing/ChartTrueEquiv.lean`.
- General common-prime input interface: `BealFinal/Main.lean`.

Standard Lean, Mathlib, and SageMath dependencies remain acknowledged. Repository-only source provenance is not a claim of having no external libraries. A `def Prop`, a folder name, or a certificate label is not by itself an inhabited theorem.

REFERENCES

- [1] DAVID FOX, *Beal Level 26 Foundations v24.4.0: v24.x Final Summary Linking Log Upper Bound to Conditional Matveev Ratio Toward B_0* . On gap-3 $A^4 + B^4 = (B + 3)^{13}$, $|\Lambda| = \log(1 + B^4/A^4) \leq B^4/A^4 = B^4/((B + 3)^{13} - B^4)$ with $0 < \exp(C_{\text{exp_bound}}) < 10^{-12} < 1$, $C_{\text{exp_bound}} = -\text{height}_{B_0} \log \text{height}_{B_0}$, $\text{height}_{B_0} = 104382751019310000000 = C_1^{\text{floor}} \cdot 30^6$. IF Matveev (2000) Thm 1.4 target $|\Lambda| > \exp(C)$ held, THEN $0 < \exp(C) < B^4/A^4$ and $B^4(1 + \exp(C)) > \exp(C)(B + 3)^{13}$ toward Baker $B_0 = 10^6$. No lower bound claimed. The historical release record reports Lean verification, 0 sorry, and `[propxt, Classical.choice, Quot.sound]` for its stated scope; this is not a fresh verification result here. DOI <https://doi.org/10.5281/zenodo.22732209>, concept <https://doi.org/10.5281/zenodo.22379293>, tag `v24.4.0-Beal-44-13-Level-26-v24x-Final-Summary` on `db7a556`.
- [2] DAVID FOX, *Beal conjecture repository: v37-true-chart*. Version DOI <https://doi.org/10.5281/zenodo.23120540>. The true-chart presentation is cited through this chart-version record and the immutable repository paths above, not as a general Beal proof certificate.